HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and executed **root cause analysis (RCA)** procedures for recurring equipment failures on critical production machinery, identifying underlying issues that led to a **15% reduction** in unscheduled downtime.
- Designed and implemented **preventive maintenance schedules** for filling lines and lyophilizers, incorporating data from sensor readings to optimize component lifespan and improve overall equipment effectiveness (OEE).
- Monitored and analyzed key equipment performance indicators including **MTBF and MTTR**, developing reports that highlighted trends and areas for proactive intervention.
- Supported **equipment qualification and validation activities** within a GMP-regulated pharmaceutical manufacturing environment, ensuring all documentation met strict regulatory compliance standards.
- Collaborated with cross-functional teams to define and implement **lean Six Sigma initiatives** aimed at enhancing process efficiency and reducing waste in the Fill and Finish production area.
- Analyzed historical performance data using **statistical methods** and predictive maintenance technologies to forecast potential equipment issues and inform maintenance strategies.
- Managed **change control processes** for equipment modifications and upgrades, ensuring all changes were documented, approved, and implemented with minimal disruption to production schedules.

SKILLS

Reliability Engineering – RCA, FMEA, RCM, MTBF, MTTR, OEE, Predictive Maintenance, Preventive Maintenance, Equipment Maintenance, Continuous Improvement

Manufacturing & Compliance – GMP, cGMP, Pharmaceutical Manufacturing, Regulatory Compliance, Validation, Equipment Qualification

Data Analysis & Tools – Data Analysis, Statistical Methods, Vibration Analysis, Thermography, Excel, SQL

Methodologies – Lean Six Sigma, Continuous Improvement, Root Cause Analysis, Failure Mode Effects Analysis

Technical Skills – Python, Java, SQL, AWS, Linux, Docker, Jenkins, Jira, Splunk, Dynatrace

PROJECTS

Equipment Reliability Enhancement Program (HCL Technologies)

- Analyzed failure data for a portfolio of enterprise client assets using **SQL and Python**, identifying key components with high failure rates and recommending targeted maintenance strategies.
- Developed automated reporting scripts in **Python** to track equipment uptime and OEE metrics, providing operations teams with near real-time visibility into asset performance.
- Contributed to the implementation of a new **predictive maintenance** sensor suite on critical machinery, integrating data feeds into an operational dashboard for proactive issue detection.

Production Bottleneck Analysis & Optimization (MSc Data Analytics)

- Conducted a comprehensive **data analysis** of a simulated manufacturing workflow, identifying bottlenecks and proposing process improvements using **Power BI** dashboards and **R** statistical analysis.
- Applied **FMEA principles** to map potential failure points in the simulated production line and quantified their impact on overall throughput.
- Presented findings and recommendations to a panel of industry experts, demonstrating a clear path to improving **equipment reliability** and reducing operational risks.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)